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►B ►M9 REGULATION (EU) No 575/2013 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL

of 26 June 2013

on prudential requirements for credit institutions and amending Regulation (EU) No 648/2012 ◄

►C2 **(Text with EEA relevance)** ◄

(OJ L 176, 27.6.2013, p. 1)

Amended by:

		Official Journal		
		No	page	date
►M1	Commission Delegated Regulation (EU) 2015/62 of 10 October 2014	L 11	37	17.1.2015
►M2	Regulation (EU) 2016/1014 of the European Parliament and of the Council of 8 June 2016	L 171	153	29.6.2016
►M3	Commission Delegated Regulation (EU) 2017/2188 of 11 August 2017	L 310	1	25.11.2017
►M4	Regulation (EU) 2017/2395 of the European Parliament and of the Council of 12 December 2017	L 345	27	27.12.2017
►M5	Regulation (EU) 2017/2401 of the European Parliament and of the Council of 12 December 2017	L 347	1	28.12.2017
►M6	Commission Delegated Regulation (EU) 2018/405 of 21 November 2017	L 74	3	16.3.2018
►M7	Regulation (EU) 2019/630 of the European Parliament and of the Council of 17 April 2019	L 111	4	25.4.2019
►M8	Regulation (EU) 2019/876 of the European Parliament and of the Council of 20 May 2019	L 150	1	7.6.2019
►M9	Regulation (EU) 2019/2033 of the European Parliament and of the Council of 27 November 2019	L 314	1	5.12.2019
►M10	Regulation (EU) 2019/2160 of the European Parliament and of the Council of 27 November 2019	L 328	1	18.12.2019
►M11	Regulation (EU) 2020/873 of the European Parliament and of the Council of 24 June 2020	L 204	4	26.6.2020
►M12	Commission Delegated Regulation (EU) 2021/424 of 17 December 2019	L 84	1	11.3.2021
►M13	Regulation (EU) 2021/558 of the European Parliament and of the Council of 31 March 2021	L 116	25	6.4.2021

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$$LGD_{floor} = LGD_{U\text{-}floor} \cdot \frac{E_U}{E \cdot (1 + H_E)} + LGD_{S\text{-}floor} \cdot \frac{E_S}{E \cdot (1 + H_E)}$$

where:

$LGD_{U\text{-}floor}$ and $LGD_{S\text{-}floor}$ are the relevant floor values in Table 1;

E, E_S, E_U and H_E are determined in accordance with Article 230.

7. Where an institution that uses own estimates of LGD for a given type of unsecured exposures to corporates and unsecured exposures to regional governments, local authorities and public sector entities is not able to take into account the effect of the funded credit protection securing one of the exposures of that type of exposures in the own estimate of LGD due to lack of data on recoveries for that funded credit protection, the institution shall be permitted to apply the formula set out in Article 230, with the exception that the LGD_U in that formula shall be the institution's own estimate of LGD for unsecured exposures. In that case, the funded credit protection shall be eligible in accordance with Chapter 4 and the institution's own estimate of LGD used as LGD_U shall be calculated based on underlying loss data excluding any recoveries arising from that funded credit protection.

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Article 162

Maturity

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1. For exposures for which an institution has not received permission from the competent authority to use own estimates of LGD, the maturity value (M) shall be applied consistently and, either be set at 2,5 years, except for exposures arising from securities financing transactions, for which M shall be 0,5 years, or, alternatively, be calculated in accordance with paragraph 2.

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2. ►M17 For exposures for which an institution applies own estimates of LGD, the maturity value (M) shall be calculated using periods expressed in years, as set out in this paragraph and subject to paragraphs 3, 4 and 5 of this Article. M shall be no greater than five years, except in the cases specified in Article 384(2) where M as specified therein shall be used. M shall be calculated as follows in each of the following cases: ◄

(a) for an instrument subject to a cash flow schedule, M shall be calculated in accordance with the following formula:

$$M = max\left\{1, min\left\{\frac{\sum_t t \cdot CF_t}{\sum_t CF_t}, 5\right\}\right\}$$

where CF_t denotes the cash flows (principal, interest payments and fees) contractually payable by the obligor in period t;

▼C2

(b) for derivatives subject to a master netting agreement, M shall be the weighted average remaining maturity of the exposure, where M shall be at least 1 year, and the notional amount of each exposure shall be used for weighting the maturity;

(c) for exposures arising from fully or nearly-fully collateralised derivative instruments listed in Annex II and fully or nearly-fully collateralised margin lending transactions which are subject to a master netting agreement, M shall be the weighted average remaining maturity of the transactions where M shall be at least 10 days;

(d) for repurchase transactions or securities or commodities lending or borrowing transactions which are subject to a master netting agreement, M shall be the weighted average remaining maturity of the transactions where M shall be at least five days. The notional amount of each transaction shall be used for weighting the maturity;

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(da) for secured lending transactions which are subject to a master netting agreement, M shall be the weighted average remaining maturity of the transactions where M shall be at least 20 days; the notional amount of each transaction shall be used for weighting the maturity;

(db) for a master netting agreement including more than one of the transaction types corresponding to point (c), (d) or (da) of this paragraph, M shall be the weighted average remaining maturity of the transactions where M shall be at least the longest holding period, expressed in years, applicable to such transactions as provided for in Article 224(2), either 10 days or 20 days, depending on the cases; the notional amount of each transaction shall be used for weighting the maturity;

▼C2

(e) an institution that has received the permission of the competent authority pursuant to Article 143 to use own PD estimates for purchased corporate receivables, for drawn amounts M shall equal the purchased receivables exposure weighted average maturity, where M shall be at least 90 days. This same value of M shall also be used for undrawn amounts under a committed purchase facility provided that the facility contains effective covenants, early amortisation triggers, or other features that protect the purchasing institution against a significant deterioration in the quality of the future receivables it is required to purchase over the facility's term. Absent such effective protections, M for undrawn amounts shall be calculated as the sum of the longest-dated potential receivable under the purchase agreement and the remaining maturity of the purchase facility, where M shall be at least 90 days;

▼M17

(f) for any instrument other than those referred to in this paragraph or when an institution is not in a position to calculate M as set out in point (a), M shall be the maximum remaining time, in years, that the obligor is permitted to take to fully discharge its contractual obligations, including the principal, interest, and fees, where M shall be at least one year;

▼C2

(g) for institutions using the Internal Model Method set out in Section 6 of Chapter 6 to calculate the exposure values, M shall be calculated for exposures to which they apply this method and for which the maturity of the longest-dated contract contained in the netting set is greater than one year in accordance with the following formula:

$$M = min\left\{\frac{\sum_k Effective\ EE_{t_k} \cdot \Delta t_k \cdot df_{t_k} \cdot s_{t_k} + \sum_k EE_{t_k} \cdot \Delta t_k \cdot df_{t_k} \cdot (1 - s_{t_k})}{\sum_k Effective\ EE_{t_k} \cdot \Delta t_k \cdot df_{t_k} \cdot s_{t_k}}, 5\right\}$$

where:

S_{t_k} = a dummy variable whose value at future period t_k is equal to 0 if $t_k > 1$ year and to 1 if $t_k \leq 1$;

EE_{t_k} = the expected exposure at the future period t_k;

$Effective\ EE_{t_k}$ = the effective expected exposure at the future period t_k;

df_{t_k} = the risk-free discount factor for future time period t_k;

Δt_k = $t_k - t_{k-1}$;

(h) an institution that uses an internal model to calculate a one-sided credit valuation adjustment (CVA) may use, subject to the permission of the competent authorities, the effective credit duration estimated by the internal model as M.

Subject to paragraph 2, for netting sets in which all contracts have an original maturity of less than one year the formula in point (a) shall apply;

▼M17

(i) for institutions using the approaches referred to in Article 382a(1), point (a) or (b), to calculate the own funds requirements for the CVA risk of transactions with a given counterparty, M shall be no greater than 1 in the formula set out in Article 153(1), point (iii), for the purpose of calculating the risk-weighted exposure amounts for counterparty risk for the same transactions, as referred to in Article 92(4), point (a) or (g), as applicable;

(j) for revolving exposures, M shall be determined using the maximum contractual termination date of the facility; institutions shall not use the repayment date of the current drawing if that date is not the maximum contractual termination date of the facility.

▼C2

3. ►M17 Where the documentation requires daily re-margining and daily revaluation and includes provisions that allow for the prompt liquidation or set off of collateral in the event of default or failure to remargin, M shall be the weighted average remaining maturity of the transactions and M shall be at least one day for: ◄

▼C2

(a) fully or nearly-fully collateralised derivative instruments listed in Annex II;

(b) fully or nearly-fully collateralised margin lending transactions;

(c) repurchase transactions, securities or commodities lending or borrowing transactions.

In addition, for qualifying short-term exposures which are not part of the institution's ongoing financing of the obligor, M shall be at least one-day. Qualifying short term exposures shall include the following:

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(a) exposures to institutions or investment firms arising from the settlement of foreign exchange obligations;

▼M17

(b) self-liquidating short-term trade finance transactions and corporate purchased receivables, provided that the respective exposures have a residual maturity of up to one year;

▼C2

(c) exposures arising from settlement of securities purchases and sales within the usual delivery period or two business days;

(d) exposures arising from cash settlements by wire transfer and settlements of electronic payment transactions and prepaid cost, including overdrafts arising from failed transactions that do not exceed a short, fixed agreed number of business days;

▼M17

(e) issued as well as confirmed letters of credit that are short term, that is, they have a maturity below one year, and are self-liquidating.

4. For exposures to corporates established in the Union which are not large corporates, institutions may choose to set for all such exposures M as set out in paragraph 1 instead of applying paragraph 2.

▼C2

5. Maturity mismatches shall be treated as specified in Chapter 4.

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6. For the purpose of expressing in years the minimum numbers of days referred to in paragraph 2, points (c) to (db), and paragraph 3, the minimum numbers of days shall be divided by 365,25.

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Sub-Section 2

Retail exposures

Article 163

Probability of default (PD)

▼M17

1. For the sole purpose of calculating risk-weighted exposure amounts and the expected loss amounts of those exposures, and in particular for the purposes of Articles 154 and 157, and Article 158 (1), (5) and (10), the PD for each exposure that is used as an input of the risk-weighted exposure amounts and expected loss formulae shall be the higher of the one-year PD associated with the internal borrower grade or pool to which the retail exposure is assigned and the following PD input floor values:

▼C2

Column 1	Column 2	Column 3	Column 4
Free delivery	No capital charge	Treat as an exposure	Treat as an exposure risk weighted at 1 250 %

2. In applying a risk weight to free delivery exposures treated according to Column 3 of Table 2, an institution using the Internal Ratings Based approach set out in Part Three, Title II, Chapter 3 may assign PDs to counterparties, for which it has no other non-trading book exposure, on the basis of the counterparty's external rating. Institutions using own estimates of 'LGDs' may apply the LGD set out in Article 161(1) to free delivery exposures treated according to Column 3 of Table 2 provided that they apply it to all such exposures. Alternatively, an institution using the Internal Ratings Based approach set out in Part Three, Title II, Chapter 3 may apply the risk weights of the Standardised Approach, as set out in Part Three, Title II, Chapter 2 provided that it applies them to all such exposures or may apply a 100 % risk weight to all such exposures.

If the amount of positive exposure resulting from free delivery transactions is not material, institutions may apply a risk weight of 100 % to these exposures, except where a risk weight of 1 250 % in accordance with Column 4 of Table 2 in paragraph 1 is required.

3. As an alternative to applying a risk weight of 1 250 % to free delivery exposures according to Column 4 of Table 2 in paragraph 1, institutions may deduct the value transferred plus the current positive exposure of those exposures from Common Equity Tier 1 items in accordance with point (k) of Article 36(1).

Article 380

Waiver

Where a system wide failure of a settlement system, a clearing system or a CCP occurs, competent authorities may waive the own funds requirements calculated as set out in Articles 378 and 379 until the situation is rectified. In this case, the failure of a counterparty to settle a trade shall not be deemed a default for purposes of credit risk.

TITLE VI

OWN FUNDS REQUIREMENTS FOR CREDIT VALUATION ADJUSTMENT RISK

Article 381

Meaning of credit valuation adjustment

For the purposes of this Title and Chapter 6 of Title II, 'credit valuation adjustment' or 'CVA' means an adjustment to the mid-market valuation of the portfolio of transactions with a counterparty. That adjustment reflects the current market value of the credit risk of the counterparty to the institution, but does not reflect the current market value of the credit risk of the institution to the counterparty.

▼M17

For the purposes of this Title, ‘CVA risk’ means the risk of losses arising from changes in the value of CVA, calculated for the portfolio of transactions with a counterparty as set out in the first paragraph, due to movements in counterparty credit spread risk factors and in other risk factors embedded in the portfolio of transactions.

▼C2

Article 382

Scope

1. An institution shall calculate the own funds requirements for CVA risk in accordance with this Title for all OTC derivative instruments in respect of all of its business activities, other than credit derivatives recognised to reduce risk-weighted exposure amounts for credit risk.

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2. An institution shall include in the calculation of own funds required by paragraph 1 securities financing transactions that are fair-valued under the accounting framework applicable to the institution where the institution’s CVA risk exposures arising from those transactions are material.

▼C2

3. Transactions with a qualifying central counterparty and a client's transactions with a clearing member, when the clearing member is acting as an intermediary between the client and a qualifying central counterparty and the transactions give rise to a trade exposure of the clearing member to the qualifying central counterparty, are excluded from the own funds requirements for CVA risk.

4. The following transactions shall be excluded from the own funds requirements for CVA risk:

(a) transactions with non-financial counterparties as defined in point (9) of Article 2 of Regulation (EU) No 648/2012, or with non-financial counterparties established in a third country, where those transactions do not exceed the clearing threshold as specified in Article 10(3) and (4) of that Regulation;

▼M19

(aa) intragroup transactions entered into with non-financial counterparties as defined in Article 2, point (9), of Regulation (EU) No 648/2012 which are part of the same group provided that all the following conditions are met:

(i) the institution and the non-financial counterparties are included in the same consolidation on a full basis and are subject to supervision on a consolidated basis in accordance with Part One, Title II, Chapter 2;

(ii) they are subject to appropriate centralised risk evaluation, measurement and control procedures; and

(iii) the non-financial counterparties are established in the Union or, if they are established in a third country, the Commission has adopted an implementing act in accordance with paragraph 4c in respect of that third country;

▼M19

(b) intragroup transactions entered into with financial counterparties, as defined in Article 2, point (8), of Regulation (EU) No 648/2012, financial institutions or ancillary services undertakings that are established in the Union or that are established in a third country that applies prudential and supervisory requirements to those financial counterparties, financial institutions or ancillary services undertakings that are at least equivalent to those applied in the Union, unless Member States adopt national law requiring the structural separation within a banking group, in which case the competent authorities may require those intragroup transactions between the structurally separated entities to be included in the own funds requirements;

▼C2

(c) transactions with counterparties referred to in point (10) of Article 2 of Regulation (EU) No 648/2012 and subject to the transitional provisions set out in Article 89(1) of that Regulation until those transitional provisions cease to apply;

(d) transactions with counterparties referred to in Article 1(4) and (5) of Regulation (EU) No 648/2012 and transactions with counterparties for which Article 114(4) and Article 115(2) of this Regulation specifies a risk weight of 0 % for exposures to those counterparties.

The exemption from the CVA risk charge for those transactions referred to in point (c) of this paragraph) which are entered into during the transitional period laid down in Article 89(1) of Regulation (EU) No 648/2012 shall apply for the length of the contract of that transaction.

In regard to point (a), where an institution ceases to be exempt through crossing the exemption threshold or due to a change in the exemption threshold, outstanding contracts shall remain exempt until the date of their maturity.

▼M17

4a. By way of derogation from paragraph 4 of this Article, an institution may choose to calculate the own funds requirements for CVA risk, using any of the approaches referred to in Article 382a(1), for the transactions that are excluded pursuant to paragraph 4 of this Article, where the institution uses eligible hedges determined in accordance with Article 386 to mitigate the CVA risk of those transactions. Institutions shall establish policies to specify the application and calculation of the own funds requirements for CVA risk for such transactions.

4b. Institutions shall report to their competent authorities the results of the calculations of the own funds requirements for CVA risk for all transactions referred to in paragraph 4 of this Article. For the purposes of that reporting requirement, institutions shall calculate the own funds requirements for CVA risk using the relevant approaches set out in Article 382a(1) that they would have used to satisfy an own funds requirement for CVA risk if those transactions were not excluded from the scope pursuant to paragraph 4 of this Article.

▼M19

4c. For the purposes of paragraph 4, points (aa) and (b), the Commission may adopt, by way of implementing acts, and subject to the examination procedure referred to in Article 464(2), a decision as to whether a third country applies prudential supervisory and regulatory requirements at least equivalent to those applied in the Union.

▼C2

5. EBA shall conduct a review by 1 January 2015 and every two years thereafter, in the light of international regulatory developments and including on potential methodologies on the calibration and thresholds for application of CVA risk charges to non-financial counterparties established in a third country.

EBA in cooperation with ESMA shall develop draft regulatory technical standards to specify the procedures for excluding transactions with non-financial counterparties established in a third country from the own funds requirement for CVA risk charge.

EBA shall submit those draft regulatory technical standards within six months of the date of the review referred to in the first subparagraph,

Power is delegated to the Commission to adopt the regulatory technical standards referred to in the second subparagraph in accordance with Articles 10 to 14 of Regulation (EU) No 1093/2010.

▼M17

6. EBA shall develop draft regulatory technical standards to specify the conditions and the criteria that institutions are to use to assess whether the CVA risk exposures arising from fair-valued securities financing transactions are material, as well as the frequency of that assessment.

EBA shall submit those draft regulatory technical standards to the Commission by 10 July 2026.

Power is delegated to the Commission to supplement this Regulation by adopting the regulatory technical standards referred to in the first subparagraph of this paragraph in accordance with Articles 10 to 14 of Regulation (EU) No 1093/2010.

Article 382a

Approaches for calculating the own funds requirements for CVA risk

1. An institution shall calculate the own funds requirements for CVA risk for all transactions referred to in Article 382 in accordance with the following approaches:

(a) the standardised approach set out in Article 383, where the institution has been granted permission by the competent authority to use that approach;

(b) the basic approach set out in Article 384;

(c) the simplified approach set out in Article 385, provided that the institution meets the conditions set out in paragraph 1 of that Article.

2. An institution shall not use the approach referred to in paragraph 1, point (c), in combination with the approach referred to in point (a) or (b) of that paragraph.

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3. An institution may use a combination of the approaches referred to in paragraph 1, points (a) and (b), to calculate the own funds requirements for CVA risk on a permanent basis for:

(a) different counterparties;

(b) different eligible netting sets with the same counterparty;

(c) different transactions of the same eligible netting set, provided that any of the conditions referred to in paragraph 5 are satisfied.

4. For the purposes of paragraph 3, point (c), institutions shall split the eligible netting set into a hypothetical netting set containing the transactions subject to the approach referred to in paragraph 1, point (a), and a hypothetical netting set containing the transactions subject to the approach referred to in paragraph 1, point (b).

5. For the purposes of paragraph 3, point (c), the conditions referred to therein shall comprise the following:

(a) the split is consistent with the treatment of the legal netting set when calculating the CVA for accounting purposes;

(b) the permission granted by competent authorities to use the approach referred to in paragraph 1, point (a), is limited to the corresponding hypothetical netting set and does not cover all transactions within the eligible netting set.

Institutions shall document how they use a combination of the approaches referred to in paragraph 1, points (a) and (b), and as set out in this paragraph, to calculate the own funds requirements for CVA risk on a permanent basis.

Article 383

Standardised approach

1. The competent authority shall grant an institution permission to calculate its own funds requirements for CVA risk for a portfolio of transactions with one or more counterparties by using the standardised approach in accordance with paragraph 3 of this Article, after having assessed whether the institution complies with the following requirements:

(a) the institution has established a distinct unit which is responsible for the institution's overall risk management and hedging of CVA risk;

(b) for each counterparty concerned, the institution has developed a regulatory CVA model to calculate the CVA of that counterparty in accordance with Article 383a;

(c) for each counterparty concerned, the institution is able to calculate, at least on a monthly basis, the sensitivities of its CVA to the risk factors concerned as determined in accordance with Article 383b;

(d) for all positions in eligible hedges recognised in accordance with Article 386 for the purpose of calculating the own funds requirements for CVA risk using the standardised approach, the institution is able to calculate, and at least on a monthly basis, the sensitivities of those positions to the relevant risk factors determined in accordance with Article 383b;

▼M17

(e) the institution has established a risk control unit that is independent from business trading units and the unit referred to in point (a) and that reports directly to the management body; that risk control unit shall be responsible for designing and implementing the standardised approach and shall produce and analyse monthly reports on the output of that approach and, moreover, the risk control unit shall assess the appropriateness of the institution's trading limits and include the results of that assessment in its monthly reports; the risk control unit shall have a sufficient number of staff with a level of skills that is appropriate to fulfil its purpose.

For the purposes of the first subparagraph, point (c), of this paragraph the sensitivity of a counterparty's CVA to a risk factor means the relative change in the value of that CVA, as a result of a change in the value of one of the relevant risk factors of that CVA, calculated using the institution's regulatory CVA model in accordance with Articles 383i and 383j.

For the purposes of the first subparagraph, point (d), of this paragraph the sensitivity of a position in an eligible hedge to a risk factor means the relative change in the value of that position, as a result of a change in the value of one of the relevant risk factors of that position, calculated using the institution's pricing model in accordance with Articles 383i and 383j.

2. For the purpose of calculating the own funds requirements for CVA risk, the following definitions apply:

(1) "risk class" means any of the following categories:

(a) interest rate risk;

(b) counterparty credit spread risk;

(c) reference credit spread risk;

(d) equity risk;

(e) commodity risk;

(f) foreign exchange risk;

(2) "CVA portfolio" means the portfolio composed of the aggregate CVA and the eligible hedges referred to in paragraph 1, point (d);

(3) "aggregate CVA" means the sum of the CVAs calculated using the regulatory CVA model for the counterparties referred to in paragraph 1, first subparagraph.

3. Institutions shall determine the own funds requirements for CVA risk using the standardised approach as the sum of the following own funds requirements calculated in accordance with Article 383b:

(a) the own funds requirements for delta risk which capture the risk of changes in the institution's CVA portfolio due to movements in the relevant non-volatility related risk factors;

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(b) the own funds requirements for vega risk which capture the risk of changes in the institution's CVA portfolio due to movements in the relevant volatility related risk factors.

Article 383a

Regulatory CVA model

1. A regulatory CVA model used for calculating the own funds requirements for CVA risk in accordance with Article 383 shall be conceptually sound, implemented with integrity, and comply with all of the following requirements:

(a) the regulatory CVA model is capable of modelling the CVA of a given counterparty, recognising netting and margin agreements at netting set level, where relevant, in accordance with this Article;

(b) the institution estimates the counterparty's probabilities of default from the counterparty credit spreads and market-consensus expected loss given default for that counterparty;

(c) the expected loss given default referred to in point (a) shall be the same as the market-consensus expected loss given default referred to in point (b), unless the institution can demonstrate that the seniority of the portfolio of transactions with that counterparty differs from the seniority of senior unsecured bonds issued by that counterparty;

(d) at each future time point, the simulated discounted future exposure of the portfolio of transactions with a counterparty is calculated with an exposure model by repricing all transactions in that portfolio, based on the simulated joint changes of the market risk factors that are material to those transactions using an appropriate number of scenarios, and discounting the prices to the date of calculation using risk-free interest rates;

(e) the regulatory CVA model is capable of modelling significant dependency between the simulated discounted future exposure of the portfolio of transactions and the counterparty credit spreads;

(f) where the transactions of the portfolio are included in a netting set subject to a margin agreement and daily mark-to-market valuation, the collateral posted and received as part of that agreement is recognised as a risk mitigant in the simulated discounted future exposure, where all of the following conditions are met:

(i) the institution determines the margin period of risk relevant for that netting set in accordance with the requirements set out in Article 285(2) and (5), and reflects that margin period in the calculation of the simulated discounted future exposure;

(ii) all applicable features of the margin agreement, including the frequency of margin calls, the type of contractually eligible collateral, the threshold amounts, the minimum transfer amounts, the independent amounts and the initial margins for both the institution and the counterparty are appropriately reflected in the calculation of the simulated discounted future exposure;

▼M17

(iii) the institution has established a collateral management unit that complies with Article 287 for all collateral recognised for calculating the own funds requirements for CVA risk using the standardised approach.

For the purposes of the first subparagraph, point (a), CVA shall have a positive sign and shall be calculated as a function of the counterparty's expected loss given default, an appropriate set of the counterparty's probabilities of default at future time points and an appropriate set of simulated discounted future exposures of the portfolio of transactions with that counterparty at future time points until the maturity of the longest transaction in that portfolio.

For the purposes of the demonstration referred to in the first subparagraph, point (c), collateral received from the counterparty shall not change the seniority of the exposure.

For the purposes of the first subparagraph, point (f)(iii), of this paragraph where the institution has already established a collateral management unit for using the internal model method referred to in Article 283, the institution shall not be required to establish an additional collateral management unit where that institution demonstrates to its competent authority that such a unit complies with the requirements set out in Article 287 for the collateral recognised for calculating the own funds requirements for CVA risk using the standardised approach.

2. For the purposes of paragraph 1, point (b), where the credit default swap spreads of the counterparty are observable in the market, an institution shall use those spreads. Where such credit default swap spreads are not available, an institution shall use one of the following:

(a) credit spreads from other instruments issued by the counterparty reflecting current market conditions;

(b) proxy spreads that are appropriate considering the rating, industry and region of the counterparty.

3. An institution using a regulatory CVA model shall comply with all of the following qualitative requirements:

(a) the exposure model referred to in paragraph 1 is part of the institution's internal CVA risk management system that includes the identification, measurement, management, approval and internal reporting of CVA and CVA risk for accounting purposes;

(b) the institution has in place a process for ensuring compliance with a documented set of internal policies, controls, assessment of model performance and procedures concerning the exposure model referred to in paragraph 1;

(c) the institution shall have an independent validation unit that is responsible for the effective initial and ongoing validation of the exposure model referred to in paragraph 1 of this Article; that unit shall be independent from business credit and trading units, including the unit referred to in Article 383(1), point (a), and report directly to senior management; it shall have a sufficient number of staff with a level of skills that is appropriate to fulfil that purpose;

▼M17

(d) the senior management shall be actively involved in the risk control process and shall regard CVA risk control as an essential aspect of the business, to which appropriate resources need to be devoted;

(e) the institution shall document the process for initial and ongoing validation of the exposure model referred to in paragraph 1 to a level of detail that would enable a third party to understand how the models operate, their limitations, and their key assumptions, and recreate the analysis; that documentation shall set out the minimum frequency with which ongoing validation will be conducted, as well as other circumstances, such as a sudden change in market behaviour, under which additional validation shall be conducted; it shall describe how the validation is conducted with respect to data flows and portfolios, what analyses are used and how representative counterparty portfolios are constructed;

(f) the pricing models used in the exposure model referred to in paragraph 1 for a given scenario of simulated market risk factors shall be tested against appropriate independent benchmarks for a wide range of market states as part of the initial and ongoing model validation process; pricing models for options shall account for the non-linearity of option value with respect to market risk factors;

(g) an independent review of the institution's internal CVA risk management system referred to in point (a) of this paragraph shall be carried out by the institution's internal auditing process on a regular basis; that review shall include the activities both of the unit referred to in Article 383(1), point (a), and of the independent validation unit referred to in point (c) of this paragraph;

(h) the regulatory CVA model used by the institution for calculating the simulated discounted future exposure referred to in paragraph 1, shall reflect transaction terms and specifications and margin agreements in a timely, complete, and conservative manner; the terms and specifications shall reside in a secure database subject to formal and periodic audit; the transmission of transaction terms and specifications data and margin agreements to the exposure model shall also be subject to internal audit, and formal reconciliation processes shall be in place between the internal model and source data systems to verify on an ongoing basis that transaction terms, specifications and margin agreements are being reflected in the exposure system correctly or, at least, conservatively;

(i) the current and historical market data inputs used in the model by the institution for calculating the simulated discounted future exposure referred to in paragraph 1 shall be acquired independently of the business lines and fed into that model in a timely and complete manner and maintained in a secure database subject to formal and periodic audit; an institution shall have a well-developed data integrity process to handle inappropriate data observations; where the model relies on proxy market data, an institution shall design internal policies to identify suitable proxies and shall demonstrate empirically on an ongoing basis that the proxies provide a conservative representation of the underlying risk;

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(j) the exposure model referred to in paragraph 1 shall capture the transaction specific and contractual information necessary in order to aggregate exposures at the level of the netting set; an institution shall verify that transactions are assigned to the appropriate netting set within the model.

For the purpose of calculating the own funds requirements for CVA risk, the exposure model referred to in paragraph 1 of this Article may have different specifications and assumptions in order to meet all requirements laid down in Article 383a, except that its market data inputs and netting recognition shall remain the same as the ones used for accounting purposes.

4. EBA shall develop draft regulatory technical standards to specify:

(a) how proxy spreads referred to in paragraph 2, point (b), are to be determined by the institution for the purposes of calculating default probabilities;

(b) further technical elements that institutions are to take into account when calculating the counterparty's expected loss given default, the counterparty's probabilities of default and the simulated discounted future exposure of the portfolio of transactions with that counterparty and CVA, as referred to in paragraph 1;

(c) which other instruments referred to in paragraph 2, point (a), are appropriate to estimate the counterparty's probabilities of default and how institutions are to make that estimate.

EBA shall submit those draft regulatory technical standards to the Commission by 10 July 2027.

Power is delegated to the Commission to supplement this Regulation by adopting the regulatory technical standards referred to in the first subparagraph of this paragraph in accordance with Articles 10 to 14 of Regulation (EU) No 1093/2010.

5. EBA shall develop draft regulatory technical standards to specify:

(a) the conditions for assessing the materiality of extensions and changes to the use of the standardised approach as referred to in Article 383(3);

(b) the assessment methodology under which competent authorities are to verify an institution's compliance with the requirements set out in Articles 383 and 383a.

EBA shall submit those draft regulatory technical standards to the Commission by 10 July 2028.

Power is delegated to the Commission to supplement this Regulation by adopting the regulatory technical standards referred to in the first subparagraph of this paragraph in accordance with Articles 10 to 14 of Regulation (EU) No 1093/2010.

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Article 383b

Own funds requirements for delta and vega risks

1. Institutions shall apply the delta and vega risk factors described in Articles 383c to 383h, and the process set out in paragraphs 2 to 8 of this Article, to calculate the own funds requirements for delta and vega risks.

2. For each risk class referred to in Article 383(2), the sensitivity of the aggregate CVAs and the sensitivity of all positions in eligible hedges falling within the scope of the own funds requirements for delta or vega risk to each of the applicable delta or vega risk factors included in that risk class shall be calculated by using the corresponding formulae set out in Articles 383i and 383j. Where the value of an instrument depends on several risk factors, the sensitivity shall be determined separately for each risk factor.

For the calculation of the vega risk sensitivities of the aggregate CVAs, sensitivities both to volatilities used in the exposure model to simulate risk factors and to volatilities used to reprice option transactions in the portfolio with the counterparty shall be included.

By way of derogation from paragraph 1 of this Article, subject to the permission of the competent authority, an institution may use alternative definitions of delta and vega risk sensitivities in the calculation of the own funds requirements of a trading book position under this Chapter, provided that the institution meets all of the following conditions:

(a) those alternative definitions are used for internal risk management purposes or for the reporting of profits and losses to senior management by an independent risk control unit within the institution;

(b) the institution demonstrates that those alternative definitions are more appropriate for capturing the sensitivities of the position than the formulae set out in Articles 383i and 383j, and that the resulting delta and vega risk sensitivities do not materially differ from the ones obtained applying the formulae set out in Articles 383i and 383j, respectively.

3. Where an eligible hedge is an index instrument, institutions shall calculate the sensitivities of that eligible hedge to all relevant risk factors by applying the shift of one of the relevant risk factors to each of the index constituents.

4. An institution may introduce additional risk factors that correspond to qualified index instruments for the following risk classes:

(a) counterparty credit spread risk;

(b) reference credit spread risk; and

(c) equity risk.

For the purposes of delta risks, an index instrument shall be considered qualified where it meets the conditions set out in Article 325i. For vega risks, all index instruments shall be considered qualified.

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An institution shall calculate sensitivities of CVA and eligible hedges to qualified index risk factors in addition to sensitivities to the non-index risk factors.

An institution shall calculate delta and vega risk sensitivities to a qualified index risk factor as a single sensitivity to the underlying qualified index. Where 75 % of the constituents of a qualified index are mapped to the same sector as set out in Articles 383p, 383s and 383v, the institution shall map the qualified index to that same sector. Otherwise, the institution shall map the sensitivity to the applicable qualified index bucket.

5. The weighted sensitivities of the aggregate CVA and of the market value of all eligible hedges to each risk factor shall be calculated by multiplying the respective net sensitivities by the corresponding risk weight, in accordance with the following formulae:

$$WS_k^{CVA} = RW_k \cdot S_k^{CVA}$$

$$WS_k^{hedges} = RW_k \cdot S_k^{hedges}$$

where:

k = the index that denotes the risk factor k;

WS_k^{CVA} = the weighted sensitivity of the aggregate CVA to risk factor k;

RW_k = the risk weight applicable to the risk factor k;

S_k^{CVA} = the net sensitivity of the aggregate CVA to risk factor k;

WS_k^{hedges} = the weighted sensitivity of the market value of all eligible hedges in the CVA portfolio to risk factor k;

S_k^{hedges} = the net sensitivity of the market value of all eligible hedges in the CVA portfolio to risk factor k.

6. Institutions shall calculate the net-weighted sensitivity WS_k of the CVA portfolio to risk factor k in accordance with the following formula:

$$WS_k = WS_k^{CVA} - WS_k^{hedges}$$

7. The net-weighted sensitivities within the same bucket shall be aggregated in accordance with the following formula, using the corresponding correlations ρ_{kl} to weighted sensitivities within the same bucket set out in Articles 383l, 383t and 383q giving rise to the bucket-specific sensitivity K_b:

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$$K_b = \sqrt{\sum_{k\epsilon b} WS_k^2 + \sum_{k\epsilon b} \sum_{l\epsilon b, k\neq l} \rho_{kl} WS_k WS_l + R \cdot \sum_{k\epsilon b} \left(\left(WS_k^{hedges} \right)^2 \right)}$$

where:

K_b = the bucket-specific sensitivity of bucket b;

WS_k = the net-weighted sensitivities;

ρ_{kl} = the corresponding intra-bucket correlation parameters;

R = the hedging disallowance parameter equal to 0,01.

8. The bucket-specific sensitivity shall be calculated in accordance with paragraphs 5, 6 and 7 of this Article for each bucket within a risk class. Once the bucket-specific sensitivity has been calculated for all buckets, weighted sensitivities to all risk factors across buckets shall be aggregated in accordance with the following formula, using the corresponding correlations γ_{bc} for weighted sensitivities in different buckets set out in Articles 383l, 383o, 383r, 383u, 383w and 383z giving rise to the risk-class specific own funds requirements for delta or vega risk:

$$risk - \text{class specific own funds requirement } for\ delta\ or\ vega\ risk = m_{CVA} \sqrt{\sum_b K_b^2 + \sum_b \sum_{b\neq c} \gamma_{bc} S_b S_c}$$

where:

m_{CVA} = a multiplier factor which is equal to 1; the competent authority may increase the value of m_{CVA} where the institution's regulatory CVA model shows deficiencies preventing the appropriate measurement of the own funds requirements for CVA risk;

K_b = the bucket-specific sensitivity of bucket b;

γ_{bc} = the correlation parameter between buckets b and c;

$S_b = max\left\{-K_b; min\left(\sum_{k\epsilon b} WS_k ; K_b\right)\right\}$ for all risk factors in bucket b;

$S_c = max\left\{-K_c; min\left(\sum_{k\epsilon c} WS_k ; K_c\right)\right\}$ for all risk factors in bucket c.

Article 383c

Interest rate risk factors

1. For the interest rate delta risk factors, including inflation rate risk, there shall be one bucket per currency, with each bucket containing different types of risk factors.

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The interest rate delta risk factors that are applicable to interest-rate sensitive instruments in the CVA portfolio shall be the risk-free rates per currency concerned and per each of the following maturities: 1 year, 2 years, 5 years, 10 years and 30 years.

The interest rate delta risk factors applicable to inflation-rate sensitive instruments in the CVA portfolio shall be the inflation rates per currency concerned and per each of the following maturities: 1 year, 2 years, 5 years, 10 years and 30 years.

2. The currencies for which an institution shall apply the interest rate delta risk factors in accordance with paragraph 1 shall be euro, Swedish krona, Australian dollar, Canadian dollar, British pound sterling, Japanese yen and US dollar, the institution's reporting currency and the currency of a Member State participating in ERM II.

3. For currencies not specified in paragraph 2, the interest rate delta risk factors shall be the absolute change of the inflation rate and the parallel shift of the entire risk-free curve for a given currency.

4. Institutions shall obtain the risk-free rates per currency from money market instruments held in their trading book that have the lowest credit risk, including overnight index swaps.

5. Where institutions cannot apply the approach referred to in paragraph 4, the risk-free rates shall be based on one or more market-implied swap curves used by the institutions to mark positions to market, such as the interbank offered rate swap curves.

Where the data on market-implied swap curves described in the first subparagraph are insufficient, the risk-free rates may be derived from the most appropriate sovereign bond curve for a given currency.

6. The interest rate vega risk factor applicable to instruments in the CVA portfolio sensitive to interest rate volatility shall be all the volatilities of the interest rate of all tenors for a given currency. The inflation rate vega risk factor applicable to instruments in the CVA portfolio sensitive to inflation rate volatility shall be all the volatilities of the inflation rate of all tenors for a given currency. There shall be one net interest rate sensitivity and one net inflation rate sensitivity computed for each currency.

Article 383d

Foreign exchange risk factors

1. The foreign exchange delta risk factors to be applied by institutions to instruments in the CVA portfolio sensitive to foreign exchange spot rates shall be the spot foreign exchange rates between the currency in which an instrument is denominated and the institution's reporting currency or the institution's base currency where the institution is using a base currency in accordance with Article 325q(7). There shall be one bucket per currency pair, containing a single risk factor and a single net sensitivity.

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2. The foreign exchange vega risk factors to be applied by institutions to instruments in the CVA portfolio sensitive to foreign exchange volatility shall be the implied volatilities of foreign exchange rates between the currency pairs referred to in paragraph 1. There shall be one bucket for all currencies and maturities, containing all foreign exchange vega risk factors and a single net sensitivity.

3. Institutions shall not be required to distinguish between onshore and offshore variants of a currency for foreign exchange delta and vega risk factors.

Article 383e

Counterparty credit spread risk factors

1. The counterparty credit spread delta risk factors applicable to counterparty credit spread sensitive instruments in the CVA portfolio shall be the credit spreads of individual counterparties and reference names and qualified indices for the following maturities: 0,5 years, 1 year, 3 years, 5 years and 10 years.

2. The counterparty credit spread risk class shall not be subject to vega risk own funds requirements.

Article 383f

Reference credit spread risk factors

1. The reference credit spread delta risk factors applicable to reference credit spread sensitive instruments in the CVA portfolio shall be the credit spreads of all maturities for all reference names within a bucket. There shall be one net sensitivity computed for each bucket.

2. The reference credit spread vega risk factors applicable to instruments in the CVA portfolio sensitive to reference credit spread volatility shall be the volatilities of the credit spreads of all tenors for all reference names within a bucket. There shall be one net sensitivity computed for each bucket.

Article 383g

Equity risk factors

1. The buckets for all equity risk factors shall be the buckets referred to in Article 383t.

2. The equity delta risk factors to be applied by institutions to instruments in the CVA portfolio sensitive to equity spot prices shall be the spot prices of all equities mapped to the same bucket referred to in paragraph 1. There shall be one net sensitivity computed for each bucket.

3. The equity vega risk factors to be applied by institutions to instruments in the CVA portfolio sensitive to equity volatility shall be the implied volatilities of all equities mapped to the same bucket referred to in paragraph 1. There shall be one net sensitivity computed for each bucket.

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Article 383h

Commodity risk factors

1. The buckets for all commodity risk factors shall be the sector buckets referred to in Article 383x.

2. The commodity delta risk factors to be applied by institutions to instruments in the CVA portfolio sensitive to commodity spot prices shall be the spot prices of all commodities mapped to the same sector bucket referred to in paragraph 1. There shall be one net sensitivity computed for each sector bucket.

3. The commodity vega risk factors to be applied by institutions to instruments in the CVA portfolio sensitive to commodity price volatility shall be the implied volatilities of all commodities mapped to the same sector bucket referred to in paragraph 1. There shall be one net sensitivity computed for each sector bucket.

Article 383i

Delta risk sensitivities

1. Institutions shall calculate delta sensitivities consisting of interest rate risk factors as follows:

(a) the delta sensitivities of the aggregate CVA to risk factors consisting of risk-free rates, as well as of an eligible hedge to those risk factors, shall be calculated as follows:

$$S_{r_{kt}}^{CVA} = \frac{V_{CVA}(r_{kt} + 0{,}0001, x, y \ldots) - V_{CVA}(r_{kt}, x, y \ldots)}{0{,}0001}$$

$$S_{r_{kt}}^{hedge_i} = \frac{V_i(r_{kt} + 0{,}0001, w, z \ldots) - V_i(r_{kt}, w, z \ldots)}{0{,}0001}$$

where:

$S_{r_{kt}}^{CVA}$ = the sensitivities of the aggregate CVA to a risk-free rate risk factor;

r_{kt} = the value of the risk-free rate risk factor k with maturity t;

V_{CVA} = the aggregate CVA calculated by the regulatory CVA model;

x,y = risk factors other than r_{kt} in V_{CVA};

$S_{r_{kt}}^{hedge_i}$ = the sensitivities of the eligible hedge i to a risk-free rate risk factor;

V_i = the pricing function of the eligible hedge i;

w,z = risk factors other than r_{kt} in the pricing function V_i;

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(b) the delta sensitivities to risk factors consisting of inflation rates as well as of an eligible hedge to those risk factors, shall be calculated as follows:

$$S^{CVA}_{infl_{kt}} = \frac{V_{CVA}(infl_{kt} + 0{,}0001, x, y \ldots) - V_{CVA}(infl_{kt}, x, y \ldots)}{0{,}0001}$$

$$S^{hedge_i}_{infl_{kt}} = \frac{V_i(infl_{kt} + 0{,}0001, w, z \ldots) - V_i(infl_{kt}, w, z \ldots)}{0{,}0001}$$

where:

$S^{CVA}_{infl_{kt}}$ = the sensitivities of the aggregate CVA to an inflation rate risk factor;

$infl_{kt}$ = the value of an inflation rate risk factor k with maturity t;

V_{CVA} = the aggregate CVA calculated by the regulatory CVA model;

x,y = risk factors other than $infl_{kt}$ in V_{CVA};

$S^{hedge_i}_{infl_{kt}}$ = the sensitivities of the eligible hedge i to an inflation rate risk factor;

V_i = the pricing function of the eligible hedge i;

w,z = risk factors other than $infl_{kt}$ in the pricing function V_i.

2. Institutions shall calculate the delta sensitivities of the aggregate CVA to risk factors consisting of foreign exchange spot rates, as well as of an eligible hedge instrument to those risk factors, as follows:

$$S^{CVA}_{FX_k} = \frac{V_{CVA}(FX_k \cdot 1{,}01, x, y \ldots) - V_{CVA}(FX_k, x, y \ldots)}{0{,}01}$$

$$S^{hedge_i}_{FX_k} = \frac{V_i(FX_k \cdot 1{,}01, w, z \ldots) - V_i(FX_k, w, z \ldots)}{0{,}01}$$

where:

$S^{CVA}_{FX_k}$ = the sensitivities of the aggregate CVA to a foreign exchange spot rate risk factor;

FX_k = the value of the foreign exchange spot rate risk factor k;

V_{CVA} = the aggregate CVA calculated by the regulatory CVA model;

x,y = risk factors other than FX_k in V_{CVA};

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$S_{FX_k}^{hedge_i}$ = the sensitivities of the eligible hedge i to a foreign exchange spot rate risk factor;

V_i = the pricing function of the eligible hedge i;

w,z = risk factors other than FX_k in the pricing function V_i.

3. Institutions shall calculate the delta sensitivities of the aggregate CVA to risk factors consisting of counterparty credit spread rates, as well as of an eligible hedge instrument to those risk factors, as follows:

$$S_{ccs_{kt}}^{CVA} = \frac{V_{CVA}(ccs_{kt} + 0{,}0001, x, y \ldots) - V_{CVA}(ccs_{kt}, x, y \ldots)}{0{,}0001}$$

$$S_{ccs_{kt}}^{hedge_i} = \frac{V_i(ccs_{kt} + 0{,}0001, w, z \ldots) - V_i(ccs_{kt}, w, z \ldots)}{0{,}0001}$$

where:

$S_{ccs_{kt}}^{CVA}$ = the sensitivities of the aggregate CVA to a counterparty credit spread rate risk factor;

ccs_{kt} = the value of the counterparty credit spread rate risk factor k at maturity t;

V_{CVA} = the aggregate CVA calculated by the regulatory CVA model;

x,y = risk factors other than ccs_{kt} in V_{CVA};

$S_{ccs_{kt}}^{hedge_i}$ = the sensitivities of the eligible hedge i to a counterparty credit spread rate risk factor;

V_i = the pricing function of the eligible hedge i;

w,z = risk factors other than ccs_{kt} in the pricing function V_i.

4. Institutions shall calculate the delta sensitivities of the aggregate CVA to risk factors consisting of reference credit spread rates, as well as of an eligible hedge instrument to those risk factors, as follows:

$$S_{rcs_{kt}}^{CVA} = \frac{V_{CVA}(rcs_{kt} + 0{,}0001, x, y \ldots) - V_{CVA}(rcs_{kt}, x, y \ldots)}{0{,}0001}$$

$$S_{rcs_{kt}}^{hedge_i} = \frac{V_i(rcs_{kt} + 0{,}0001, w, z \ldots) - V_i(rcs_{kt}, w, z \ldots)}{0{,}0001}$$

where:

$S_{rcs_{kt}}^{CVA}$ = the sensitivities of the aggregate CVA to a reference credit spread rate risk factor;

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rcs_{kt} = the value of the reference credit spread rate risk factor k at maturity t;

V_{CVA} = the aggregate CVA calculated by the regulatory CVA model;

x,y = risk factors other than ccs_{kt} in V_{CVA} ;

$S_{rcs_{kt}}^{hedge_i}$ = the sensitivities of the eligible hedge i to a reference credit spread rate risk factor;

V_i = the pricing function of the eligible hedge i;

w,z = risk factors other than ccs_{kt} in the pricing function V_i.

5. Institutions shall calculate the delta sensitivities of the aggregate CVA to risk factors consisting of equity spot prices, as well as of an eligible hedge instrument to those risk factors, as follows:

$$S_{EQ}^{CVA} = \frac{V_{CVA}(EQ \cdot 1{,}01, x, y \ldots) - V_{CVA}(EQ, x, y \ldots)}{0{,}01}$$

$$S_{EQ}^{hedge_i} = \frac{V_i(EQ \cdot 1{,}01, w, z \ldots) - V_i(EQ, w, z \ldots)}{0{,}01}$$

where:

S_{EQ}^{CVA} = the sensitivities of the aggregate CVA to an equity spot price risk factor;

EQ = the value of the equity spot price;

V_{CVA} = the aggregate CVA calculated by the regulatory CVA model;

x,y = risk factors other than EQ in V_{CVA};

$S_{EQ}^{hedge_i}$ = the sensitivities of the eligible hedge i to an equity spot price risk factor;

V_i = the pricing function of the eligible hedge i;

w,z = risk factors other than EQ in the pricing function V_i.

6. Institutions shall calculate the delta sensitivities of the aggregate CVA to risk factors consisting of commodity spot prices, as well as of an eligible hedge instrument to those risk factors, as follows:

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$$S_{CTY}^{CVA} = \frac{V_{CVA}(1{,}01 \cdot CTY, x, y \ldots) - V_{CVA}(CTY, x, y \ldots)}{0{,}01}$$

$$S_{CTY}^{hedge_i} = \frac{V_i(1{,}01 \cdot CTY, w, z \ldots) - V_i(CTY, w, z \ldots)}{0{,}01}$$

where:

S_{CTY}^{CVA} = the sensitivities of the aggregate CVA to a commodity spot price risk factor;

CTY = the value of the commodity spot price;

V_{CVA} = the aggregate CVA calculated by the regulatory CVA model;

x,y = risk factors other than CTY in V_{CVA};

$S_{CTY}^{hedge_i}$ = the sensitivities of the eligible hedge i to a commodity spot price risk factor;

V_i = the pricing function of the eligible hedge i;

w,z = risk factors other than CTY in the pricing function V_i.

Article 383j

Vega risk sensitivities

Institutions shall calculate the vega risk sensitivities of the aggregate CVA to risk factors consisting of implied volatility, as well as of an eligible hedge instrument to those risk factors, as follows:

$$S_{vol_k}^{CVA} = \frac{V_{CVA}(vol_k \cdot 1{,}01, x, y \ldots) - V_{CVA}(vol_k, x, y \ldots)}{0{,}01}$$

$$S_{vol_k}^{hedge_i} = \frac{V_i(vol_k \cdot 1{,}01, w, z \ldots) - V_i(vol_k, w, z \ldots)}{0{,}01}$$

where:

$S_{vol_k}^{CVA}$ = the sensitivities of the aggregate CVA to an implied volatility risk factor;

vol_k = the value of the implied volatility risk factor;

V_{CVA} = the aggregate CVA calculated by the regulatory CVA model;

x,y = risk factors other than vol_k in the pricing function V_{CVA};

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$S_{vol_k}^{hedge_i}$ = the sensitivities of the eligible hedge instrument i to an implied volatility risk factor;

V_i = the pricing function of the eligible hedge i;

w,z = risk factors other than vol_k in the pricing function V_i.

Article 383k

Risk weights for interest rate risk

1. For the currencies referred to in Article 383c(2), the risk weights of risk-free rate delta sensitivities for each bucket in Table 1 shall be the following:

Table 1

Bucket	Maturity	Risk weight
1	1 year	1,11 %
2	2 years	0,93 %
3	5 years	0,74 %
4	10 years	0,74 %
5	30 years	0,74 %

2. For currencies other than the currencies referred to in Article 383c (2), the risk weight of risk-free rate delta sensitivities shall be 1,58 %.

3. For inflation rate risk denominated in one of the currencies referred to in Article 383c(2), the risk weight of the delta sensitivity to the inflation rate risk shall be 1,11 %.

4. For inflation rate risk denominated in a currency other than the currencies referred to in Article 383c(2), the risk weight of the delta sensitivity to the inflation rate risk shall be 1,58 %.

5. The risk weights to be applied to sensitivities to interest rate vega risk factors and to inflation rate vega risk factors for all currencies shall be 100 %.

Article 383l

Intra-bucket correlations for interest rate risk

1. For the currencies referred to in Article 383c(2), the correlation parameters that institutions shall apply to the aggregation of the risk-free rate delta sensitivities between the different buckets set out in Article 383k, Table 1, shall be the following:

Table 1

Bucket	1	2	3	4	5
1	100 %	91 %	72 %	55 %	31 %
2		100 %	87 %	72 %	45 %

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3			100 %	91 %	68 %
4				100 %	83 %
5					100 %

2. Institutions shall apply a correlation parameter of 40 % for the aggregation of inflation rate delta risk sensitivity and risk-free rate delta sensitivity denominated in the same currency.

3. Institutions shall apply a correlation parameter of 40 % for the aggregation of inflation rate vega risk factor sensitivity and interest rate vega risk factor sensitivity denominated in the same currency.

Article 383m

Correlation across buckets for interest rate risk

The cross-bucket correlation parameter for interest rate delta and vega risks shall be set at 0,5 for all currency pairs.

Article 383n

Risk weights for foreign exchange risk

1. The risk weights for all delta sensitivities to foreign exchange risk factor between an institution's reporting currency and another currency shall be 11 %.

2. The risk weight of the foreign exchange risk factors concerning currency pairs which are composed of the euro and the currency of a Member State participating in ERM II shall be one of the following:

(a) the risk weight referred to in paragraph 1, divided by 3;

(b) the maximum fluctuation within the fluctuation band formally agreed by the Member State and the ECB, if that fluctuation band is narrower than the fluctuation band defined under ERM II.

3. Notwithstanding paragraph 2, the risk weight of the foreign exchange risk factors concerning currencies referred to in that paragraph which participate in ERM II with a formally agreed fluctuation band narrower than the standard band of plus or minus 15 % shall equal the maximum percentage fluctuation within that narrower band.

4. The risk weights for all vega sensitivities to foreign exchange risk factor shall be 100 %.

Article 383o

Correlations for foreign exchange risk

1. A uniform correlation parameter equal to 60 % shall apply to the aggregation of sensitivities to delta foreign exchange risk factor across buckets.

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2. A uniform correlation parameter equal to 60 % shall apply to the aggregation of sensitivities to vega foreign exchange risk factor across buckets.

Article 383p

Risk weights for counterparty credit spread risk

1. The risk weights for the delta sensitivities to counterparty credit spread risk factors shall be the same for all maturities (0,5 years, 1 year, 3 years, 5 years, 10 years) within each bucket in Table 1 and shall be the following:

Table 1

Bucket number	Credit quality	Sector	Risk weight
1	All	Central government, including central banks, of Member States	0,5 %
2	Credit quality step 1 to 3	Central government, including central banks, of third countries, multilateral development banks and international organisations referred to in Article 117(2) and Article 118	0,5 %
3		Regional government or local authority and public sector entities	1,0 %
4		Financial sector entities, including credit institutions incorporated or established by a central government, a regional government or a local authority, and promotional lenders	5,0 %
5		Basic materials, energy, industrials, agriculture, manufacturing, mining and quarrying	3,0 %
6		Consumer goods and services, transportation and storage, administrative and support service activities	3,0 %
7		Technology, telecommunications	2,0 %
8		Health care, utilities, professional and technical activities	1,5 %
9		Covered bonds issued by credit institutions established in Member States	1,0 %
10	Credit quality step 1	Covered bonds issued by credit institutions in third countries	1,5 %
	Credit quality steps 2 to 3		2,5 %
11	Credit quality steps 1 to 3	Other sector	5,0 %
12		Qualified indices	1,5 %

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Bucket number	Credit quality	Sector	Risk weight
13	Credit quality step 4 to 6 and unrated	Central government, including central banks, of third countries, multilateral development banks and international organisations referred to in Article 117(2) and Article 118	2,0 %
14		Regional government or local authority and public sector entities	4,0 %
15		Financial sector entities, including credit institutions incorporated or established by a central government, a regional government or a local authority, and promotional lenders	12,0 %
16		Basic materials, energy, industrials, agriculture, manufacturing, mining and quarrying	7,0 %
17		Consumer goods and services, transportation and storage, administrative and support service activities	8,5 %
18		Technology, telecommunications	5,5 %
19		Health care, utilities, professional and technical activities	5,0 %
20		Other sector	12,0 %
21		Qualified indices	5,0 %

Where there are no external ratings for a specific counterparty, institutions may, subject to approval by the competent authorities, map the internal rating to a corresponding external rating and assign a risk weight corresponding to either credit quality step 1 to 3 or credit quality step 4 to 6. Otherwise, the risk weights for unrated exposures shall be applied.

2. To assign a risk exposure to a sector, institutions shall rely on a classification that is commonly used in the market for grouping issuers by sector. Institutions shall assign each issuer to only one of the sector buckets set out in Table 1. Risk exposures from any issuer that an institution cannot assign to a sector in such a manner shall be assigned to either bucket 11 or bucket 20 in Table 1, depending on the credit quality of the issuer.

3. Institutions shall assign to buckets 12 and 21 in Table 1 only exposures that reference qualified indices as referred to in Article 383b(4).

4. Institutions shall use a look-through approach to determine the sensitivities of an exposure referencing a non-qualified index.

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Article 383q

Intra-bucket correlations for counterparty credit spread risk

1. Between two sensitivities WS_k and WS_l, resulting from risk exposures assigned to sector buckets 1 to 11 and 13 to 20, as set out in Article 383p(1), Table 1, the correlation parameter ρ_{kl} shall be set as follows:

$$\rho_{kl} = \rho_{kl}^{(tenor)} \cdot \rho_{kl}^{(name)} \cdot \rho_{kl}^{(quality)}$$

where:

$\rho_{kl}^{(tenor)}$ shall be equal to 1 where the two vertices of the sensitivities k and l are identical, otherwise it shall be equal to 90 %;

$\rho_{kl}^{(name)}$ shall be equal to 1 where the two names of sensitivities k and l are identical, 90 % if the two names are distinct but legally related, otherwise it shall be equal to 50 %;

$\rho_{kl}^{(quality)}$ shall be equal to 1 where the two names are both in buckets 1 to 11 or are both in buckets 13 to 20, otherwise it shall be equal to 80 %.

2. Between two sensitivities WS_k and WS_l resulting from risk exposures assigned to sector buckets 12 and 21, the correlation parameter ρ_{kl} shall be set as follows:

$$\rho_{kl} = \rho_{kl}^{(tenor)} \cdot \rho_{kl}^{(name)} \cdot \rho_{kl}^{(quality)}$$

where:

$\rho_{kl}^{(tenor)}$ shall be equal to 1 where the two vertices of the sensitivities k and l are identical, otherwise it shall be equal to 90 %;

$\rho_{kl}^{(name)}$ shall be equal to 1 where the two names of sensitivities k and l are identical and the two indices are of the same series, 90 % if the two indices are the same but of distinct series, otherwise it shall be equal to 80 %;

$\rho_{kl}^{(quality)}$ shall be equal to 1 where the two names are both in bucket 12 or both in bucket 21, otherwise it shall be equal to 80 %.

Article 383r

Correlations across buckets for counterparty credit spread risk

The cross-bucket correlations for counterparty credit spread delta risk shall be the following:

▼M17

Table 1

Bucket	1, 2, 3, 13 and 14	4 and 15	5 and 16	6 and 17	7 and 18	8 and 19	9 and 10	11 and 20	12 and 21
1, 2, 3, 13 and 14	100 %	10 %	20 %	25 %	20 %	15 %	10 %	0 %	45 %
4 and 15		100 %	5 %	15 %	20 %	5 %	20 %	0 %	45 %
5 and 16			100 %	20 %	25 %	5 %	5 %	0 %	45 %
6 and 17				100 %	25 %	5 %	15 %	0 %	45 %
7 and 18					100 %	5 %	20 %	0 %	45 %
8 and 19						100 %	5 %	0 %	45 %
9 and 10							100 %	0 %	45 %
11 and 20								100 %	0 %
12 and 21									100 %

Article 383s

Risk weights for reference credit spread risk

1. The risk weights for the delta sensitivities to reference credit spread risk factors shall be the same for all maturities (0,5 years, 1 year, 3 years, 5 years, 10 years) and all reference credit spread exposures within each bucket in Table 1 and shall be the following:

Table 1

Bucket number	Credit quality	Sector	Risk weight
1	All	Central government, including central banks, of Member States	0,5 %
2	Credit quality step 1 to 3	Central government, including central banks, of third countries, multilateral development banks and international organisations referred to in Article 117(2) and Article 118	0,5 %
3		Regional government or local authority and public sector entities	1,0 %
4		Financial sector entities, including credit institutions incorporated or established by a central government, a regional government or a local authority, and promotional lenders	5,0 %
5		Basic materials, energy, industrials, agriculture, manufacturing, mining and quarrying	3,0 %
6		Consumer goods and services, transportation and storage, administrative and support service activities	3,0 %
7		Technology, telecommunications	2,0 %
8		Health care, utilities, professional and technical activities	1,5 %
9		Covered bonds issued by credit institutions established in Member States	1,0 %

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Bucket number	Credit quality	Sector	Risk weight
10	Credit quality step 1	Covered bonds issued by credit institutions in third countries	1,5 %
	Credit quality steps 2 to 3		2,5 %
11	Credit Quality Step 1 to 3	Qualified indices	1,5 %
12	Credit quality step 4 to 6 and unrated	Central government, including central banks, of third countries, multilateral development banks and international organisations referred to in Article 117(2) and Article 118	2,0 %
13		Regional government or local authority and public sector entities	4,0 %
14		Financial sector entities, including credit institutions incorporated or established by a central government, a regional government or a local authority, and promotional lenders	12,0 %
15		Basic materials, energy, industrials, agriculture, manufacturing, mining and quarrying	7,0 %
16		Consumer goods and services, transportation and storage, administrative and support service activities	8,5 %
17		Technology, telecommunications	5,5 %
18		Health care, utilities, professional and technical activities	5,0 %
19		Qualified indices	5,0 %
20	Other sector		12,0 %

Where there are no external ratings for a specific counterparty, institutions may, subject to approval by the competent authorities, map the internal rating to a corresponding external rating and assign a risk weight corresponding to either credit quality step 1 to 3 or credit quality step 4 to 6. Otherwise, the risk weights for unrated exposures shall be applied.

2. Risk weights for reference credit spread volatilities shall be set at 100 %.

3. To assign a risk exposure to a sector, institutions shall rely on a classification that is commonly used in the market for grouping issuers by sector. Institutions shall assign each issuer to only one of the sector buckets in Table 1. Risk exposures from any issuer that an institution cannot assign to a sector in such a manner shall be assigned to bucket 20 in Table 1.

4. Institutions shall assign to buckets 11 and 19 only exposures that reference qualified indices as referred to in Article 383b(4).

5. Institutions shall use a look-through approach to determine the sensitivities of an exposure referencing a non-qualified index.

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Article 383t

Intra-bucket correlations for reference credit spread risk

1. Between two sensitivities WS_k and WS_l, resulting from risk exposures assigned to sector buckets 1 to 10, 12 to 18 and 20 of Article 383s(1), Table 1, the correlation parameter ρ_{kl} shall be set as follows:

$$\rho_{kl} = \rho_{kl}^{(tenor)} \cdot \rho_{kl}^{(name)} \cdot \rho_{kl}^{(quality)}$$

where:

$\rho_{kl}^{(tenor)}$ shall be equal to 1 where the two vertices of the sensitivities k and l are identical, otherwise it shall be equal to 90 %;

$\rho_{kl}^{(name)}$ shall be equal to 1 where the two names of sensitivities k and l are identical, 90 % if the two names are distinct but legally related, otherwise it shall be equal to 50 %;

$\rho_{kl}^{(quality)}$ shall be equal to 1 where the two names are both in buckets 1 to 10, are both in buckets 12 to 18, or are both in bucket 20, otherwise it shall be equal to 80 %.

2. Between two sensitivities WS_k and WS_l, resulting from risk exposures assigned to sector buckets 11 and 19, the correlation parameter ρ_{kl} shall be set as follows:

$$\rho_{kl} = \rho_{kl}^{(tenor)} \cdot \rho_{kl}^{(name)} \cdot \rho_{kl}^{(quality)}$$

where:

$\rho_{kl}^{(tenor)}$ shall be equal to 1 where the two vertices of the sensitivities k and l are identical, otherwise it shall be equal to 90 %;

$\rho_{kl}^{(name)}$ shall be equal to 1 where the two names of sensitivities k and l are identical and the two indices are of the same series, 90 % if the two indices are the same but of distinct series, otherwise it shall be equal to 80 %;

$\rho_{kl}^{(quality)}$ shall be equal to 1 where the two names are both in bucket 11 or both in bucket 19, otherwise it shall be equal to 80 %.

Article 383u

Correlations across buckets for reference credit spread risk

1. The cross-bucket correlations for reference credit spread delta risk and reference credit spread vega risk shall be the following:

▼**M17**

Table 1

Bucket	1, 2 and 12	3 and 14	4 and 15	5 and 16	6 and 17	7 and 18	8 and 19	9 and 10	20	11	19
1, 2, and 12	100 %	75 %	10 %	20 %	25 %	20 %	15 %	10 %	0 %	45 %	45 %
3 and 14		100 %	5 %	15 %	20 %	15 %	10 %	10 %	0 %	45 %	45 %
4 and 15			100 %	5 %	15 %	20 %	5 %	20 %	0 %	45 %	45 %
5 and 16				100 %	20 %	25 %	5 %	5 %	0 %	45 %	45 %
6 and 17					100 %	25 %	5 %	15 %	0 %	45 %	45 %
7 and 18						100 %	5 %	20 %	0 %	45 %	45 %
8 and 19							100 %	5 %	0 %	45 %	45 %
9 and 10								100 %	0 %	45 %	45 %
20									100 %	0 %	0 %
11										100 %	75 %
19											100 %

2. By way of derogation from paragraph 1, the cross-bucket correlation values calculated in that paragraph shall be divided by 2 for correlations between a bucket from the group of buckets 1 to 10 and a bucket from the group of buckets 12 to 18.

Article 383v

Risk weight buckets for equity risk

1. The risk weights for the delta sensitivities to equity spot price risk factors shall be the same for all equity risk exposures within each bucket in Table 1 and shall be the following:

Table 1

Bucket number	Market capitalisation	Economy	Sector	Risk weight for equity spot price
1	Large	Emerging market economy	Consumer goods and services, transportation and storage, administrative and support service activities, healthcare, utilities	55 %
2			Telecommunications, industrials	60 %
3			Basic materials, energy, agriculture, manufacturing, mining and quarrying	45 %
4			Financials, including government-backed financials, immovable property activities, technology	55 %
5		Advanced economy	Consumer goods and services, transportation and storage, administrative and support service activities, healthcare, utilities	30 %
6			Telecommunications, industrials	35 %
7			Basic materials, energy, agriculture, manufacturing, mining and quarrying	40 %
8			Financials, including government-backed financials, immovable property activities, technology	50 %

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Bucket number	Market capitalisation	Economy	Sector	Risk weight for equity spot price
9	Small	Emerging market economy	All sectors described under bucket numbers 1, 2, 3 and 4	70 %
10		Advanced economy	All sectors described under bucket numbers 5, 6, 7 and 8	50 %
11	Other sector			70 %
12	Large	Advanced economy	Qualified indices	15 %
13	Other		Qualified indices	25 %

2. For the purposes of paragraph 1 of this Article, what constitutes a small and a large capitalisation shall be specified in the regulatory technical standards referred to in Article 325bd(7).

3. For the purposes of paragraph 1 of this Article, what constitutes an emerging market and an advanced economy shall be specified in the regulatory technical standards referred to in Article 325ap(3).

4. When assigning a risk exposure to a sector, institutions shall rely on a classification that is commonly used in the market for grouping issuers by industry sector. Institutions shall assign each issuer to one of the sector buckets in paragraph 1, Table 1, and shall assign all issuers from the same industry to the same sector. Risk exposures from any issuer that an institution cannot assign to a sector in that manner shall be assigned to bucket 11. Multinational or multi-sector equity issuers shall be allocated to a particular bucket on the basis of the most material region and sector in which the equity issuer operates.

5. The risk weights for equity vega risk shall be set at 78 % for buckets 1 to 8 and bucket 12, and at 100 % for all other buckets.

Article 383w

Correlations across buckets for equity risk

The cross-bucket correlation parameter for equity delta and vega risk shall be set at:

(a) 15 %, where the two buckets fall within buckets 1 to 10 in Article 383v(1), Table 1;

(b) 75 %, where the two buckets are buckets 12 and 13 in Article 383v (1), Table 1;

(c) 45 %, where one of the buckets is bucket 12 or 13 in Article 383v (1), Table 1, and the other bucket falls within buckets 1 to 10 in Article 383v(1), Table 1;

(d) 0 %, where one of the two buckets is bucket 11 in Article 383v(1), Table 1.

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Article 383x

Risk weight buckets for commodity risk

1. The risk weights for the delta sensitivities to commodity spot price risk factors shall be the same for all commodity risk exposures within each bucket in Table 1 and shall be the following:

Table 1

Bucket number	Bucket name	Risk weight for commodity spot price
1	Energy — solid combustibles	30 %
2	Energy — liquid combustibles	35 %
3	Energy — electricity	60 %
4	Energy — EU ETS carbon trading	40 %
5	Energy — non-EU ETS carbon trading	60 %
6	Freight	80 %
7	Metals — non-precious	40 %
8	Gaseous combustibles	45 %
9	Precious metals, including gold	20 %
10	Grains and oilseed	35 %
11	Livestock and dairy	25 %
12	Softs and other agricultural commodities	35 %
13	Other commodity	50 %

2. The risk weights for commodity vega risk shall be set at 100 %.

Article 383z

Correlations across buckets for commodity risk

1. The cross-bucket correlation parameter for commodity delta risk shall be set at:

(a) 20 %, where the two buckets fall within buckets 1 to 12 in Article 383x(1), Table 1;

(b) 0 %, where one of the two buckets is bucket 13 in Article 383x(1), Table 1.

2. The cross-bucket correlation parameter for commodity vega risk shall be set at:

(a) 20 %, where the two buckets fall within buckets 1 to 12 in Article 383x(1), Table 1;

(b) 0 %, where one of the two buckets is bucket 13 in Article 383x(1), Table 1.

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Article 384

Basic approach

1. An institution shall calculate the own funds requirements for CVA risk in accordance with paragraph 2 or 3 of this Article, as applicable, for a portfolio of transactions with one or more counterparties by using one of the following formulae, as appropriate:

(a) the formula set out in paragraph 2 of this Article, where the institution includes in the calculation one or more eligible hedges recognised in accordance with Article 386;

(b) the formula set out in paragraph 3 of this Article, where the institution does not include in the calculation any eligible hedges recognised in accordance with Article 386.

The approaches set out in the first subparagraph, points (a) and (b), shall not be used in combination.

2. An institution that meets the condition referred to in paragraph 1, point (a), shall calculate the own funds requirements for CVA risk as follows:

$$BACVA^{total} = \beta \cdot BACVA^{csr-unhedged} + DS_{CVA} \cdot (1 - \beta) \cdot BACVA^{csr-hedged}$$

where:

$BACVA^{total}$ = the own funds requirements for CVA risk under the basic approach;

$BACVA^{csr-unhedged}$ = the own funds requirements for CVA risk under the basic approach as calculated in accordance with paragraph 3 for an institution that meets the condition set out in paragraph 1, point (b);

DS_{CVA} = 0,65;

β = 0,25;

$$BACVA^{csr-hedged} = \sqrt{\left(\rho \cdot \sum_c (SCVA_c - SNH_c) - IH\right)^2 + (1 - \rho^2) \cdot \sum_c (SCVA_c - SNH_c)^2 + \sum_c HMA_c}$$

where:

$$SCVA_c = \frac{1}{a} \cdot RW_c \cdot \sum_{NS \in c} M_{NS}^c \cdot EAD_{NS}^c \cdot DF_{NS}^c$$

$$SNH_c = \sum_{h \in c} r_{hc} \cdot RW_h^{SN} \cdot M_h^{SN} \cdot B_h^{SN} \cdot DF_h^{SN}$$

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$$IH = \sum_i RW_i^{ind} \cdot M_i^{ind} \cdot B_i^{ind} \cdot DF_i^{ind}$$

$$HMA_c = \sum_h (1 - r_{hc}^2) \cdot (RW_h \cdot M_h^{SN} \cdot B_h^{SN} \cdot DF_h^{SN})^2$$

α = 1,4;

ρ = 0,5;

c = the index that denotes all counterparties for which the institution calculates the own funds requirements for CVA risk using the approach laid down in this Article;

NS = the index that denotes all netting sets with a given counterparty for which the institution calculates the own funds requirements for CVA risk using the approach laid down in this Article;

h = the index that denotes all single-name instruments recognised as eligible hedges in accordance with Article 386 for a given counterparty for which the institution calculates the own funds requirements for CVA risk using the approach laid down in this Article;

I = the index that denotes all index instruments recognised as eligible hedges in accordance with Article 386 for all counterparties for which the institution calculates the own funds requirements for CVA risk using the approach laid down in this Article;

RW_c = the risk weight applicable to counterparty c; counterparty c shall be mapped to one of the risk weights based on a combination of sector and credit quality and determined in accordance with Table 1.

Where there are no external ratings for a specific counterparty, institutions may, subject to approval by the competent authorities, map the internal rating to a corresponding external rating and assign a risk weight corresponding to either credit quality step 1 to 3 or credit quality step 4 to 6; otherwise, the risk weights for unrated exposures shall be applied.

M_{NS}^c= the effective maturity for the netting set NS with counterparty c;

M_{NS}^c shall be calculated in accordance with Article 162; however, for that calculation, M_{NS}^c shall not be capped at five years, but at the longest contractual remaining maturity in the netting set;

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EAD_{NS}^{c} = the counterparty credit risk exposure value of the netting set NS with counterparty c, including the effect of collateral in accordance with the methods set out in Title II, Chapter 6, Sections 3 to 6, as applicable to the calculation of the own funds requirements for counterparty credit risk referred to in Article 92(4), points (a) and (g);

DF_{NS}^{c} = the supervisory discount factor for the netting set NS with counterparty c.

For an institution, using the methods set out in Title II, Chapter 6, Section 6, the supervisory discount factor shall be set at 1; in all other cases, the supervisory discount factor shall be calculated as follows:

$$\frac{1 - e^{-0,05 \cdot M_{NS}^{c}}}{0,05 \cdot M_{NS}^{c}}$$

r_{hc} = the supervisory correlation factor between the credit spread risk of counterparty c and the credit spread risk of a single-name instrument recognised as an eligible hedge h for counterparty c, determined in accordance with Table 2;

M_{h}^{SN} = the residual maturity of a single-name instrument recognised as an eligible hedge;

B_{h}^{SN} = the notional of a single name instrument recognised as an eligible hedge;

DF_{h}^{SN} = the supervisory discount factor for a single name instrument recognised as an eligible hedge, calculated as follows:

$$\frac{1 - e^{-0,05 \cdot M_{h}^{SN}}}{0,05 \cdot M_{h}^{SN}}$$

RW_{h}^{SN} = the supervisory risk weight of a single-name instrument recognised as an eligible hedge; those risk weights shall be based on a combination of sector and credit quality of the reference credit spread of the hedging instrument and determined in accordance with Table 1;

M_{i}^{ind} = the residual maturity of one or more positions in the same index instrument recognised as an eligible hedge; in the case of more than one position in the same index instrument, M_{i}^{ind} shall be the notional-weighted maturity of all those positions;

B_{i}^{ind} = the full notional of one or more positions in the same index instrument recognised as an eligible hedge;

DF_{i}^{ind} = the supervisory discount factor for one or more positions in the same index instrument recognised as an eligible hedge, calculated as follows:

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$$\frac{1 - e^{-0,05 \cdot M_i^{ind}}}{0,05 \cdot M_i^{ind}}$$

RW_i^{ind} = the supervisory risk weight of an index instrument recognised as an eligible hedge; RW_i^{ind} shall be based on a combination of sector and credit quality of all index constituents, calculated as follows:

(a) where all index constituents belong to the same sector and have the same credit quality, as determined in accordance with Table 1, RW_i^{ind} shall be calculated as the relevant risk weight of Table 1 for that sector and credit quality multiplied by 0,7;

(b) where all index constituents do not belong to the same sector or do not have the same credit quality, RW_i^{ind} shall be calculated as a weighted average of the risk weights of all index constituents, as determined in accordance with Table 1, multiplied by 0,7;

Table 1

Sector of counterparty	Credit quality	
	Credit quality step 1 to 3	Credit quality step 4 to 6 and not rated
Central government, including central banks, multilateral development banks and international organisations referred to in Article 117(2) or Article 118	0,5 %	2,0 %
Regional government or local authority and public sector entities	1,0 %	4,0 %
Financial sector entities, including credit institutions incorporated or established by a central government, a regional government or a local authority, and promotional lenders	5,0 %	12,0 %
Basic materials, energy, industrials, agriculture, manufacturing, mining and quarrying	3,0 %	7,0 %
Consumer goods and services, transportation and storage, administrative and support service activities	3,0 %	8,5 %
Technology, telecommunications	2,0 %	5,5 %
Health care, utilities, professional and technical activities	1,5 %	5,0 %
Other sector	5,0 %	12,0 %

Table 2

Correlations between credit spread of counterparty and single-name hedge	
Single-name hedge h of counterparty i	Value of r_{hc}
Counterparties referred to in Article 386(3), point (a)(i)	100 %
Counterparties referred to in Article 386(3), point (a)(ii)	80 %
Counterparties referred to in Article 386(3), point (a)(iii)	50 %

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3. An institution that meets the condition referred to in paragraph 1, point (b), shall calculate the own funds requirements for CVA risk as follows:

$$BACVA^{csr-unhedged} = DS_{CVA} \cdot \sqrt{\left(\rho \cdot \sum_c SCVA_c\right)^2 + (1-\rho^2) \cdot \sum_c SCVA_c^2}$$

where all of the terms are the ones set out in paragraph 2.

Article 385

Simplified approach

1. An institution that meets all of the conditions set out in Article 273a(2) or has been permitted by its competent authority in accordance with Article 273a(4) to apply the approach set out in Article 282, may calculate the own funds requirements for CVA risk as the risk-weighted exposure amounts for counterparty risk for non-trading book and trading book positions, respectively, referred to in Article 92(4), points (a) and (g), divided by 12,5.

2. For the purposes of the calculation referred to in paragraph 1, the following requirements shall apply:

(a) only transactions subject to the own funds requirements for CVA risk laid down in Article 382 are subject to that calculation;

(b) credit derivatives that are recognised as internal hedges against counterparty risk exposures are not included in that calculation.

3. An institution that no longer meets one or more of the conditions set out in Article 273a(2) or (4), as applicable, shall comply with the requirements set out in Article 273b.

Article 386

Eligible hedges

1. Positions in hedging instruments shall be recognised as eligible hedges for the calculation of the own funds requirements for CVA risk in accordance with Articles 383 and 384 where those positions meet all of the following requirements:

(a) they are used for the purpose of mitigating CVA risk and are managed as such;

(b) they can be entered into with third parties or with the institution's trading book as an internal hedge, in which case they are to comply with Article 106(7);

(c) only positions in hedging instruments as referred to in paragraphs 2 and 3 of this Article can be recognised as eligible hedges for the calculation of the own funds requirements for CVA risk in accordance with Articles 383 and 384, respectively.

▼M17

For the purpose of calculating the own funds requirements for CVA risk in accordance with Article 383, positions in hedging instruments shall be recognised as eligible hedges where, in addition to the conditions set out in points (a) to (c) of this paragraph, such hedging instruments form a single position in an eligible hedge and are not split into more than one position in more than one eligible hedge.

2. For the calculation of the own funds requirements for CVA risk in accordance with Article 383, only positions in the following hedging instruments shall be recognised as eligible hedges:

(a) instruments that hedge variability of the counterparty credit spread, with the exception of instruments referred to in Article 325(5);

(b) instruments that hedge variability of the exposure component of CVA risk, with the exception of the instruments referred to in Article 325(5).

3. For the calculation of the own funds requirements for CVA risk in accordance with Article 384, only positions in the following hedging instruments shall be recognised as eligible hedges:

(a) single-name credit default swaps and single-name contingent-credit default swaps, referencing:

(i) the counterparty directly;

(ii) an entity legally related to the counterparty, where legally related refers to cases where the reference name and the counterparty are either a parent undertaking and its subsidiary or two subsidiaries of a common parent;

(iii) an entity that belongs to the same sector and region as the counterparty;

(b) index credit default swaps.

4. Positions in hedging instruments entered into with third parties that are recognised as eligible hedges in accordance with paragraphs 1, 2 and 3 and included in the calculation of the own funds requirements for CVA risk shall not be subject to the own funds requirements for market risk set out in Title IV.

5. Positions in hedging instruments that are not recognised as eligible hedges in accordance with this Article shall be subject to the own funds requirements for market risk set out in Title IV.

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PART FOUR

LARGE EXPOSURES

Article 387

Subject matter

Institutions shall monitor and control their large exposures in accordance with this Part.